

ERP Documentation – 1. Academics Department

Version 0.1 (draft) · Scope: roles, features, workflows, approvals, permissions, integrations

1. Purpose and Scope

The Academics department is fully focused on education. It covers teaching, learning, assessment, student records and academic administration. It also **raises requests to, and receives requests from, every other department** (store, finance, HR, transport, IT, housekeeping, etc.).

Users in this department: Director of Academics (DA), Principal, Coordinator, Teacher, Student, PRM (Front Office), Librarian.

Cross-department roles that interact with Academics: MD (views everything), HR, Finance, Store, Transport Manager, Security, IT Support.

2. Roles and Hierarchy

2.1 Hierarchy (approval authority, high to low)

MD → Director of Academics (DA) → Principal → Coordinator → Teacher → Student

PRM (Front Office) and Librarian are service roles that sit alongside the academic hierarchy and report into it.

2.2 Role responsibilities

Role	Primary responsibility
Director of Academics (DA)	Final academic authority. Approves timetables, academic calendar, lesson plans (final level), exam schedules, marks publishing. Reassigns teachers in emergencies.
Principal	Creates timetable, exam timetable and academic calendar (office of the principal). Approves lesson plans, gate passes, TC requests. Handles escalated complaints. Alters teacher schedules during sudden leave.

Role	Primary responsibility
Coordinator	Supervises 2 or more classes along with their class teachers. Has all teacher rights plus supervisory duties: approves lesson plans, reviews marks, monitors attendance and student performance across supervised classes.
Teacher	Supervises/teaches 1 or 2 classes . Class Teacher = class mentor (maintains student information, attendance, class-level approvals). Subject Teacher = handles specific subjects. Creates lesson plans, question banks, homework, marks.
Student	Views own profile, timetable, marks, attendance. Submits HomeFun, applies for events and activities, requests TC, raises complaints.
PRM (Front Office)	Visitor and admission-enquiry management, appointments, gate pass / product pass requests, student enrolment in extra activities, class-section change request. Can view full student details.
Librarian	Book catalogue, members, issue/return, announcements.

2.3 Teacher vs Coordinator

Aspect	Teacher	Coordinator
Classes handled	1-2	2 or more (with class teachers)
Teaching rights	Yes	Yes (same rights as teacher)
Supervision	Own class	All supervised classes
Approval power	Class-level	Lesson plan and request approvals
Own timetable + supervised timetables	Own only	Both

3. Module Map

#	Module	Primary owner
1	Dashboard and Analytics	All users
2	User Information System (UIS)	Class Teacher (students) / HR + HOD (staff)
3	Timetable (class, teacher, exam)	Principal → DA
4	Academic Calendar, Events and Announcements	Principal → DA
5	Lesson Plan	Teacher
6	Exams, Question Bank, Marks and Results	Teacher / Coordinator
7	HomeFun (Homework and Assignments)	Teacher / Student
8	Attendance (RFID-based)	Automatic + Class Teacher
9	Leave and Substitution	HOD + HR
10	Star Rating	Higher rank → lower rank
11	Task Management	All departments
12	Communication (Messaging)	All users
13	Complaints and Escalation	All users
14	Tickets and Requests (Store, IT, Housekeeping)	All users
15	Gate Pass	PRM / Class Teacher
16	TC Request and NOC	Student → MD
17	Study Material	Teacher
18	Student Evaluation and Performance Analytics	Teacher / Coordinator
19	Extra-Curricular Activities	PRM
20	Class / Section Change	Class Teacher / PRM
21	Library	Librarian
22	PRM (Front Office)	PRM
23	Notifications	System

#	Module	Primary owner
24	General Settings and Global Rules	System Admin

4. Module Details and Workflows

4.1 Dashboard and Analytics

- Every user has a role-based dashboard with graphs and charts.
- Clicking any chart opens the detailed underlying data.
- Examples: Student – marks trend, attendance %; Teacher – class performance, pending approvals; Coordinator – multi-class comparison; Principal / DA – school-wide academics; MD – all departments, comparisons and audit data.

4.2 User Information System (UIS)

A lifetime record for every user. A student's record runs from Grade 1 to Grade 10 (and beyond) without breaks.

Student record contains:

- Basic information, parent details, sibling information (sibling in same school is displayed)
- Uploaded documents, certificates, achievements
- Academic history: marks, exams, activities, extra classes
- Attendance history
- Fee details: paid / pending, concessions (sibling, staff-child, special cases)
- Transport: school bus (with vehicle details) or own transport
- Complaints, disciplinary and other activity history

Staff record contains: basic info, documents, activity history, achievements.

Who updates:

- Student data → Class Teacher (class mentor)
- Staff data → HR and the respective Department Head

Who can view:

- Own profile → the user
- Student profile → their teachers and the inter-department chain (Teacher, Coordinator, Principal, DA, MD, PRM)

- Staff profile → respective department head chain, HR, MD

Rule: Once entered, data cannot be edited or deleted (see 4.24).

4.3 Timetable

Types: Class timetable, Teacher timetable, Coordinator view (supervised classes), Exam timetable.

Flow:

1. Principal creates timetable / exam timetable
2. DA approves
3. Published → notification sent → appears in timetable window of all relevant users

Views: Teachers see own and class timetable. Coordinators see own plus all supervised classes. Students see class timetable.

4.4 Academic Calendar, Events and Announcements

- Academic calendar created by the Principal's office, approved by DA
- Event scheduling and announcements (school events, tournaments, inter-school competitions)
- Students can apply if willing to participate
- Lost-and-found announcements (photo upload by finder)
- Notifications target selected audiences

4.5 Lesson Plan

Flow:

1. Teacher creates monthly lesson plan
2. Coordinator approves
3. Principal approves
4. DA approves → final

Rules:

- Last submission date: **25th of every month** (for the next month)
- Auto-reminder **7 days before** the deadline
- Teachers can also view approval status at each level

4.6 Exams, Question Bank, Marks and Results

Components: Exam scheduling, question bank, key notes, marks entry (single and bulk upload), results.

Flow:

1. Exam schedule created by Principal → approved by DA → notification
2. Teachers upload question banks / key notes
3. Teacher enters marks (manual or bulk)
4. Higher ranks approve (Coordinator → Principal → DA)
5. Once all approve → **published**
6. Published results visible to students, parents and relevant staff

Evaluation options: CA / RCT or custom evaluation types.

4.7 HomeFun (Homework and Assignments)

- Teacher assigns homework / assignment
- Student uploads completed work
- Teacher / respective persons view submissions and enter marks
- Visible to the teacher chain

4.8 Attendance (RFID-based)

- Every user (student and staff) has an **RFID card** and punches at the device.
- Attendance is **automatically captured** from the existing attendance server and synced to the ERP (device server: iClock).
- Synced to the respective Teacher and higher ranks.

Rules:

- No punch = automatically **Absent**
- Class Teacher / Coordinator can mark Present / Absent / Leave / OD
- If a leave request exists and is approved, the status auto-updates to **Leave**
- Confirmation by the Department Head or after a defined time limit finalises the record; an auto-message can notify parents
- Same process for staff
- Users can view their own attendance

4.9 Leave and Substitution

Flow:

1. Staff submits leave request
2. Approved or denied by **HR and the respective Department Head**
3. Attendance auto-marks as Leave

Emergency / sudden leave: Principal or DA reassigns (alters) the teacher schedule to cover the absent teacher's periods.

4.10 Star Rating

- Centralised module across the school
- Higher-rank users give stars to lower ranks (students receive stars from their class teacher and subject teachers)
- **13–15 categories**; selectable for student, teaching staff or non-teaching staff
- Used to compute the **Star of the Month** by highest rank
- Results notified to everyone

4.11 Task Management

- Assign and receive tasks across **all departments**, not only within Academics
- Status tracking, assigner and assignee view, notifications

4.12 Communication (Messaging)

- One-to-one **text-only** messaging between all users
- Higher ranks can view conversations of those under them
- **MD can view all messages**

4.13 Complaints and Escalation

- Anyone (student, parent, teacher, any user) can raise a complaint
- **Resolution or action within 24 hours**
- If not resolved, or the complainant is not satisfied → **escalates to the next higher authority** (manual escalation by complainant, or automatic after 24 hours)
- Example: parent complaint about a teacher → Coordinator → Principal → DA → MD
- Full history is retained

4.14 Tickets and Requests (Store, IT, Housekeeping)

Tickets: Raise a ticket to any department (e.g., PC in repair → IT; cleaning → Housekeeping). Select the department and the person.

Store requests (stationery, supplies):

1. User views available products and counts
2. Raises request (select department and approver)
3. DA / Principal approves
4. Store approves and issues
5. User collects
6. If the product is not in stock → logged as **new product request** (feeds purchase workflow)

Budget: Finance defines monthly / yearly / event budgets in advance; requests sync against them.

4.15 Gate Pass

Flow:

1. Requested by PRM (or from student profile)
2. Selected Class Teacher approves
3. Higher ranks approve in order
4. Gate pass generated with details
5. Goes to Security profile (or printed and approved at the gate)

Related passes: Product-out pass, product-entry pass (enter here → routed to the respective department for approval).

4.16 TC Request and NOC

Flow:

1. Student (via profile) raises TC request with description
2. NOC approvals from all ranked users: Class Teacher → Coordinator → Principal → DA
3. Department NOCs: Transport Manager (if using school bus), Library, Finance (dues), others as required
4. **MD gives final approval** → TC generated

4.17 Study Material

- Teachers upload and manage study material by class
- Includes question banks and key notes
- Students view by class and subject

4.18 Student Evaluation and Performance Analytics

- View students by class with details

- Class high marks, ranks (auto-calculated)
- Evaluation by CA / RCT or custom options
- Performance analytics for student, class, subject and school

4.19 Extra-Curricular Activities

- PRM enrolls students in additional activities (badminton, taekwondo, etc.)
- Fees collected by Finance
- Monthly fee reminders

4.20 Class / Section Change

- Class Teacher or PRM can initiate (e.g., Section A → B)
- Requires approval of all higher authorities

4.21 Library

- Manage books with details, categories, issued-book records, members, announcements
- Borrowed books appear in the **borrower's profile** and in the librarian view
- Book purchase requests use dropdown-fetched data (no re-entry)
- Library NOC feeds the TC process

4.22 PRM (Front Office)

- Visitor list and admission enquiry (via call or direct visit), with follow-ups and mode of contact
- Track enquiry → campus visit → request form → admission
- Record reason if enquiry did not convert
- Appointment booking (notifies the selected person)
- View all staff details and full student details, class information, attendance
- Gate pass, product-out and product-entry requests
- **New user creation:** when a student pays the admission payment, a new user is auto-created; a temporary password is mailed and the user is **forced to change it on first login**

4.23 Notifications

- Sent on approvals, publications, announcements, deadlines, complaints, tickets
- Sender selects the target audience; recipients are notified accordingly

4.24 General Settings and Global Rules

- **No edit and no delete** for any user once data is saved (wrong entries cannot be modified by the user)
- **Filters and export** (PDF / Excel, with column selection) on every page for every user
- **Profile switch** option for users with multiple roles
- **Audit trail:** login details, entry details, page download and print/download logs visible to MD
- **Concessions:** staff-child concession, sibling concession, special cases (shown in the student record)
- All modules are interconnected and role-driven

5. Approval Chains Summary

Process	Initiator	Approval chain	Output
Timetable / Exam timetable	Principal	DA	Published + notified
Academic calendar	Principal's office	DA	Published
Lesson plan	Teacher	Coordinator → Principal → DA	Approved plan
Marks	Teacher	Coordinator → Principal → DA	Published results
Leave	Staff	HOD + HR	Attendance marked Leave
Gate pass	PRM / Student	Class Teacher → higher ranks	Pass to Security
TC	Student	All ranks + dept NOCs → MD	TC generated
Class change	Class Teacher / PRM	All higher authorities	Section updated
Store request	Any user	DA / Principal → Store	Issue to user
Complaint	Any user	Receiver (24h) → next higher rank	Resolution / escalation

6. Role-Permission Matrix (draft)

Legend: C = Create, V = View, A = Approve, U = Update (initial entry only; no edit/delete after save)

Module	Student	Teacher	Coordinator	Principal	DA	PRM	Librarian	MD
Timetable	V	V	V	C	A	V	V	V
Lesson plan	-	C	A	A	A	-	-	V
Marks	V	C	A	A	A	-	-	V
HomeFun	C (submit)	C	V	V	V	-	-	V
Attendance	V	U	U	V	V	V	-	V
Leave	C	C	A (HOD)	A	A	C	C	V
Gate pass	-	A	A	A	A	C	-	V
TC	C	A	A	A	A	V	A (NOC)	A (final)
Library	V	V	V	V	V	V	C	V
Messages	C	C	C	C	C	C	C	V (all)
Star rating	V	C	C	C	C	-	-	V
Student profile	V (own)	U (class teacher)	V	V	V	V	-	V

7. Integrations With Other Departments

Department	Interaction
HR	Staff records, leave approval, interview panels (academic staff take interviews and submit feedback), appraisals
Finance	Fees, concessions, budgets, extra-activity fees, TC dues NOC, WhatsApp bill delivery
Transport	Bus details in student record, TC NOC, pickup/drop tracking
Store	Product requests, availability view
IT Support	Tickets
Housekeeping / Maintenance	Tickets, duty allotment

Department	Interaction
Security	Gate pass entry and exit
Library	Borrow records, NOC
MD	Full visibility, final approvals, audit

8. Confirmed Decisions

Updates from review. These supersede the earlier draft wording in sections 4 and 6 where they differ.

1. **Star rating categories:** Student categories and staff categories are different. Both sets are **dynamic and configured by the DA** (add, rename or disable categories). Categories are not fixed in the system.
2. **Marks approval:** Same chain as lesson plan: Teacher → Coordinator → Principal → DA, then published.
3. **Complaint escalation: Fully automatic.** If a complaint is not solved or not replied to within 24 hours, it escalates to the next higher position. Separately, if it was resolved but the complainant clicks **Not Satisfied**, it also escalates to the higher position. Full history is retained.
4. **Corrections:** No one can change or delete saved data, and there is **no correction workflow**. Entry screens must therefore use validation, confirmation and a preview step before final submit.
5. **Parent access:** Via the **student profile only**. No separate parent login.
6. **Gate pass approvers:** Chosen by the requester (PRM) at request time. The PRM selects who must approve; the chain is not fixed.
7. **Attendance finalisation:** Automatic finalisation after a time limit (exact cutoff time to be set in settings).
8. **Evaluation categories:** Evaluation is by **examination type** (e.g., Midterm, CA, RCT). Each type is a category and types are configurable.
9. **Extra-curricular activities:** Badminton, dance, karate, rifle shooting and more. Each activity has its **own separate coach**. The activity list and coaches are managed by PRM.
10. **Library rules: 1 book per student, 5 books per staff.** A user who wants to buy a book submits a request from the portal and it is reviewed. The library also maintains a **missing books list** and a **book inventory**.

9. Additional Decisions and Recommendations

Confirmed

- **Attendance timings:** Student and staff timings are different, and each department has its own timings. Cutoff times for automatic Present/Absent finalisation are **configurable per user group and per department** in settings (no single fixed time).
- **Permission request:** A separate request type for short absence, limited to a **maximum of 2 hours** per request. Approved by the Department Head (and HR for staff). On approval, attendance shows as **Permission** rather than Absent or Leave.
- **Library duration:** Borrow duration and fine rules are **custom-configurable** (by user type) in library settings. Fixed limits remain 1 book per student and 5 per staff.

Recommendation: exam weightage

Make weightage configurable by the DA per class and per exam type, with the final result computed automatically. Suggested pattern, editable at any time before an exam cycle starts:

Exam type	Suggested weightage
CA / RCT (continuous assessment)	20%
Midterm	30%
Final / Annual	50%

Why: weightage is a school policy that changes by grade (for example, lower grades lean more on continuous assessment), so it should be a setting rather than fixed in code. Once an exam cycle starts, the weightage is locked, which fits the no-edit rule.

Next department to document (suggested order): Non-Academics (Joint Director, Stores, Transport, Housekeeping/Maintenance, IT, Camera Room, File Room) → HR → Finance → MD → Admin → Audit.